Even & Odd Numbers

Number World Study Notes

Divisibility by 2, pairing intuition, and parity rules

Every whole number falls into one of two families: even or odd. This simple classification, based on divisibility by 2, shows up everywhere in mathematics — from patterns and puzzles to algebra and number theory. This chapter covers the definitions, quick tests, and handy rules for combining even and odd numbers.

Key Definitions

Even Number

A number that is exactly divisible by 2, with no remainder. It can always be written in the form $2n$, where n is a whole number. Even numbers end in 0, 2, 4, 6, or 8.

Odd Number

A number that is NOT exactly divisible by 2; dividing by 2 leaves a remainder of 1. It can be written in the form $2n + 1$. Odd numbers end in 1, 3, 5, 7, or 9.

Pairing Intuition

If you can arrange a group of objects into pairs with nothing left over, the count is even. If exactly one object is left unpaired, the count is odd.

Rules for Sums and Products

- Even + Even = Even (e.g., $4 + 6 = 10$)
- Odd + Odd = Even (e.g., $5 + 7 = 12$)
- Even + Odd = Odd (e.g., $4 + 5 = 9$)
- Even × Even = Even, Even × Odd = Even, Odd × Odd = Odd
- 0 is considered an even number, since $0 = 2 \times 0$.
- To check if a large number is even or odd, look only at its last (ones) digit.

Worked Examples

Example 1: Is 234 even or odd?

Check the last digit: 4.

4 is one of 0, 2, 4, 6, 8, so the number is even.

Answer: Even

Example 2: What is the parity (even/odd) of 7 + 15?

7 is odd and 15 is odd.

Odd + Odd = Even.

Check: 7 + 15 = 22, which is even.

Answer: Even (22)

Example 3: Is the product of 6 and 9 even or odd?

6 is even, 9 is odd.

Even × Odd = Even.

Check: 6 × 9 = 54, which is even.

Answer: Even (54)

Example 4: Find two odd numbers whose product is 143.

Try factor pairs of 143: 143 = 11 × 13.

Both 11 and 13 are odd numbers.

Answer: 11 and 13

Practice Questions

1. Classify each as even or odd: 81, 128, 999, 1000.
2. Is the sum of two odd numbers always even? Prove it with an example and with algebra (2n+1 form).
3. If a number is even, is its square always even? Give a reason.
4. Find two odd numbers whose product is 187.
5. State True or False: '0 is an even number.'

Answer Key

1. 81 - odd, 128 - even, 999 - odd, 1000 - even.
2. Yes. Example: 5 + 7 = 12 (even). Algebra: (2n+1) + (2m+1) = 2n + 2m + 2 = 2(n+m+1), which is even.
3. Yes. Even × Even = Even, so (even)^2 is always even. Example: 6^2 = 36.
4. 11 and 17, since 11 x 17 = 187.
5. True, since 0 = 2 x 0, it fits the definition of an even number.